# CHAPTER 8

## OBJECT-ORIENTED CONCEPTS AND PRINCIPLES



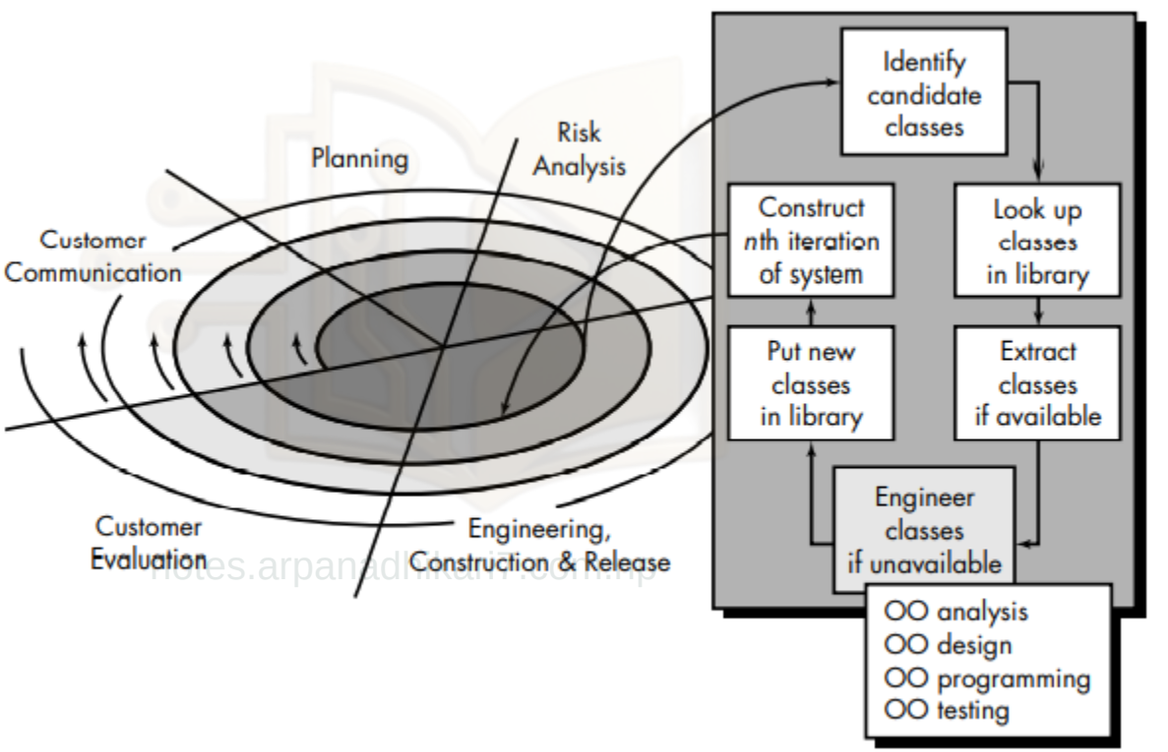
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# The OO Process model

**FIGURE 20.1**
The OO process model






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# OO Process model

- The OO process moves through an **evolutionary spiral** that starts with customer communication.
- .Planning and risk analysis establish a foundation for the OO project plan
- The technical work associated with OO software engineering follows the **iterative path** shown in the shaded box.
- OO software engineering emphasizes reuse.

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# OO Process model

- Therefore, classes are “looked up” in a library (of existing OO classes) before they are built.
- When a class cannot be found in the library, the software engineer applies object-oriented analysis (OOA), object-oriented design (OOD), object-oriented programming (OOP), and object-oriented testing (OOT) to create the class and the objects derived from the class.
- The new class is then put into the library so that it may be reused in the future

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# Definition of Object and Class

- **Class** : A class is a generalized description that describes a collection of a similar objects.
- **Object** : Instance of Class is called object

- One widely used approach to problem solving takes an **object-oriented viewpoint**

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# The Object Oriented Paradigm

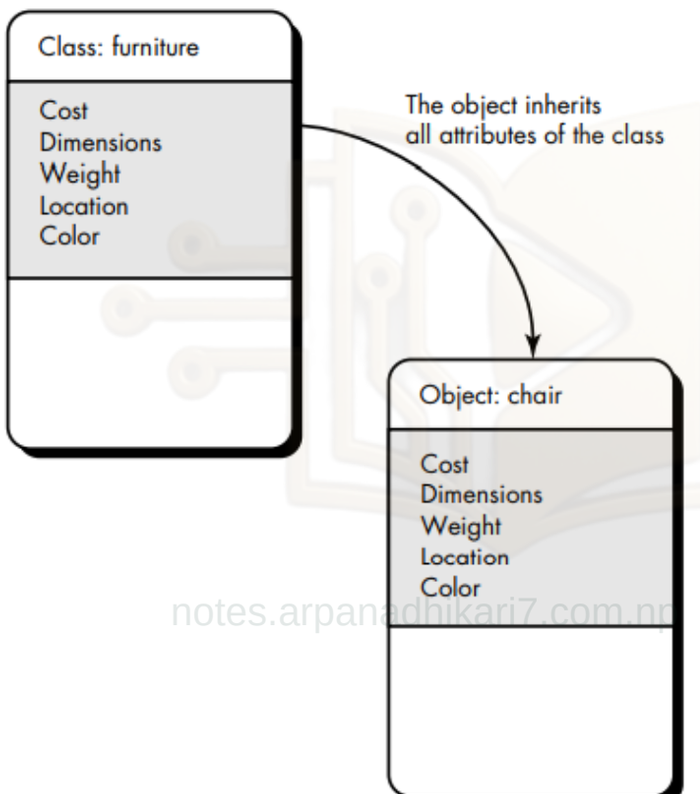
- For many years, the term object oriented (OO) was used to denote a **software development approach** that used one of a number of object-oriented programming languages Java, C++,
- Today, the OO paradigm encompasses a complete view of software engineering.

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# Inheritance

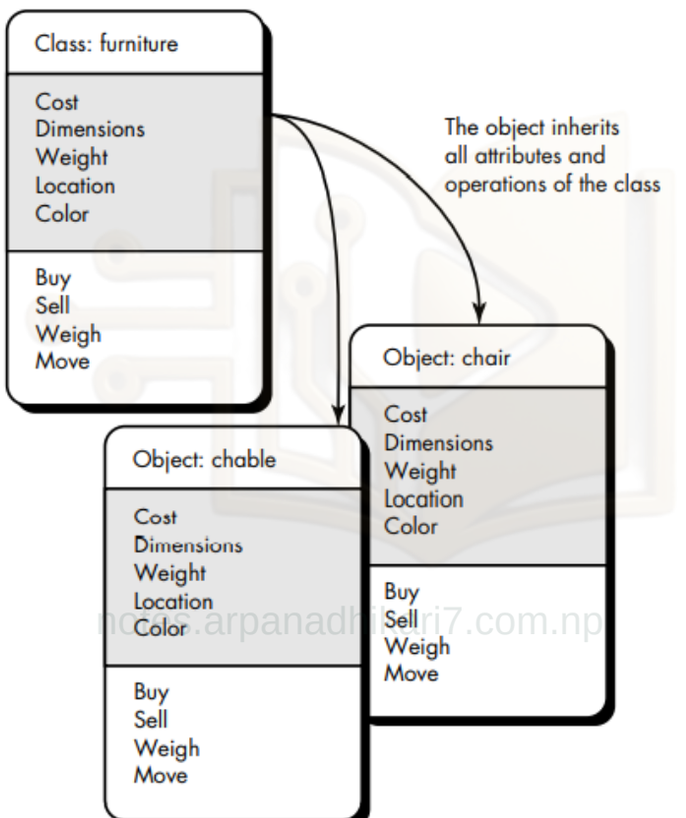



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# Inheritance

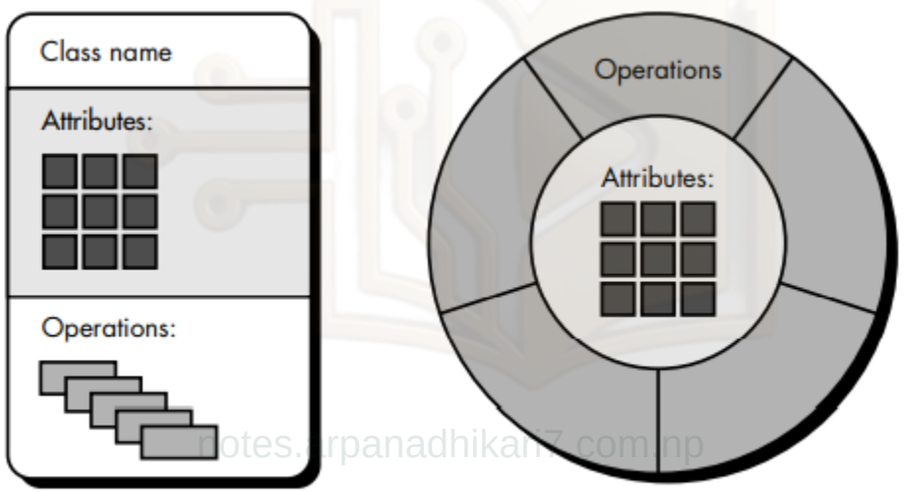





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# Class and Object

An Alternative way to represent class and object

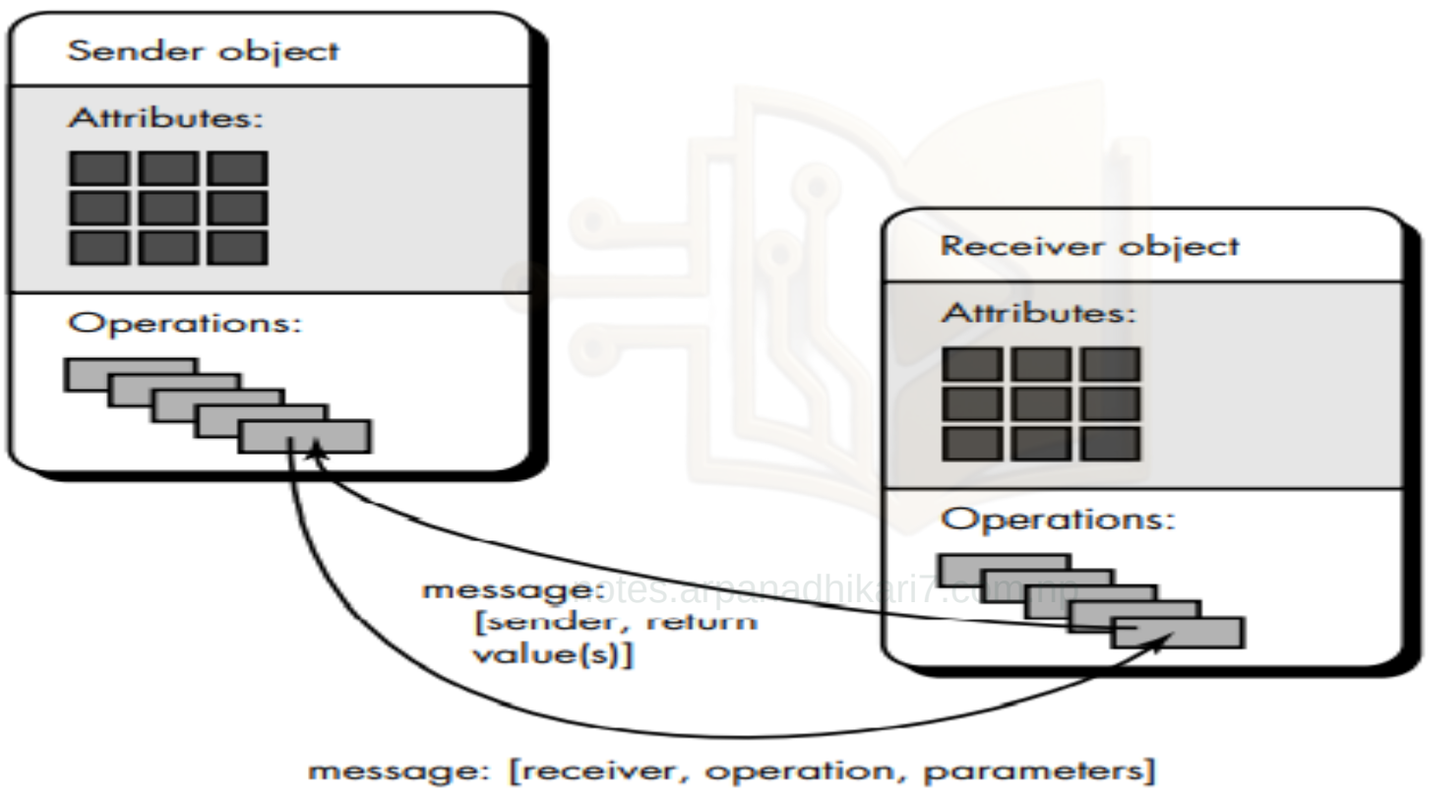



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# Message Passing Between Class






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# IDENTIFYING THE ELEMENTS OF AN OBJECT MODEL

1. **Identifying Classes and Objects**

**Objects can be:-**

- **External Entities**
- **Things**
- **Occurrences or events**
- **Roles**
- **Organizational units**
- **Places**
- **Structure**

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# IDENTIFYING THE ELEMENTS OF AN OBJECT MODE

## 2. Specifying Attributes

- Attributes are chosen by examining the problem statement, looking for things that fully define an object and make it unique.

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# IDENTIFYING THE ELEMENTS OF AN OBJECT MODE

### 3. **Defining the operation**

Although many different types of operations exist, they can generally be divided into three broad categories:

(1) operations that manipulate data in some way (e.g., adding, deleting, reformatting, selecting),

(2) operations that perform a computation,

(3) operations that monitor an object for the occurrence of a controlling event



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# IDENTIFYING THE ELEMENTS OF AN OBJECT MODE

### 3. Finalizing the Object Definition

The definition of operations is the last step in completing the specification of an object

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# OO Project Metrics and Estimation

- Conventional software project estimation -- (LOC) and function points (FP)

OOP Project Metrics is based on:

1. **Number of scenario scripts** (initiator, action and participator)
2. **Number of key classes**
3. **Number of support classes**
4. **Average number of support classes per key class**
5. **Number of subsystems.**

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